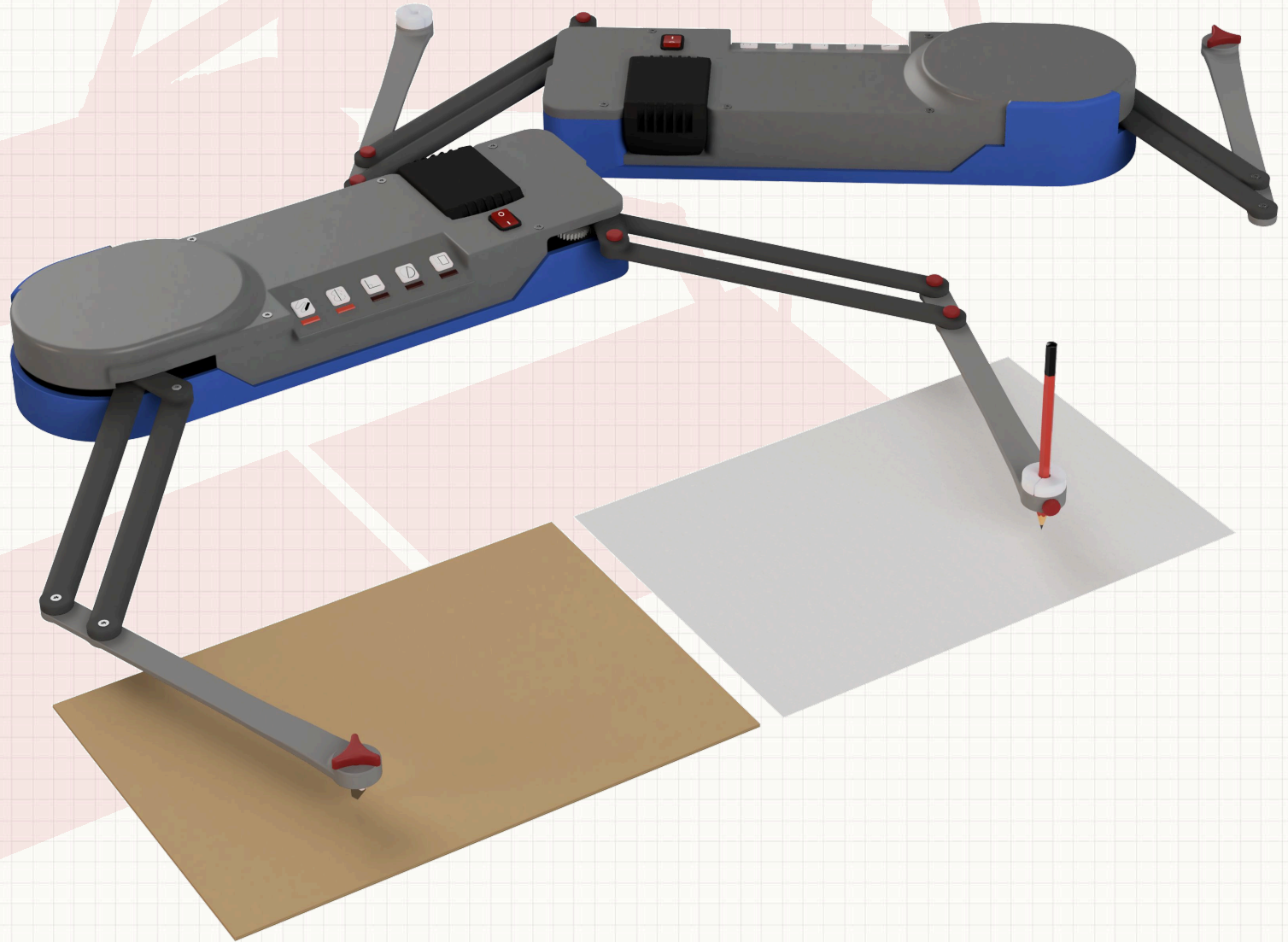


Perfect Pantograph



Ruaridh Murdoch
Individual

Individual Focus

Ruaridh Murdoch

Further Development of Opportunity

Goal and Direction

Primary Goal

Group Aim

Encourage students to utilise lo-fi prototyping stage to iterate, explore and evaluate ideas better, providing better practical hand-on technical skills

Students often struggle to turn sketches into accurate physical models. Cutting cardboard or foam with craft knives requires a lot of manual control, and small mistakes can ruin parts. This often discourages experimentation and iteration.

Design Direction

Individual Aim

My focus is improving the transition from sketch to prototype, concentrating on cutting and forming processes that reduce user error while still allowing students to control the shape they create.

Design Considerations

Safe for Classroom Use

Workshop tools already cause many minor injuries. Survey results showed 65.2% of students reported cuts or burns. Any solution should reduce exposed blades, slipping, and burn risks so it can be safely used by students aged 14–18.

Enable Student Independence

Teachers cannot assist every student immediately during lessons. Tools should work without teacher setup so students can continue modelling without waiting.

Quick and Simple to Use

DT lessons are short. Tools with long setup times reduce making time. The solution should be quick to start, easy to learn, and easy to reset between users.

Reduce Manual Skill Required

Accurate cutting currently depends on manual knife control. This makes curves and complex shapes difficult. The design should reduce the level of precision required from the user.

Clean and Manageable in Workshops

Foam modelling produces dust and debris which teachers identified as messy and a potential safety issue. A solution should contain or reduce waste during use.



Cardboard Model - Bro Edern



Cardboard Model - CLSG



Foam Model - Bro Edern

Technical Considerations

Foam Fume Management

Heated foam cutting produces harmful fumes. Most systems vent outside, which limits where the tool can be used. Adding built-in filtration allows safe indoor use in classrooms.

Battery Compatibility

Use standard cordless drill batteries (~12–20V) that schools already have. This allows quick swaps and no separate power unit. The device can sit directly on the battery or have it slot in, with voltage stepped down for electronics and motors.



Concept Exploration

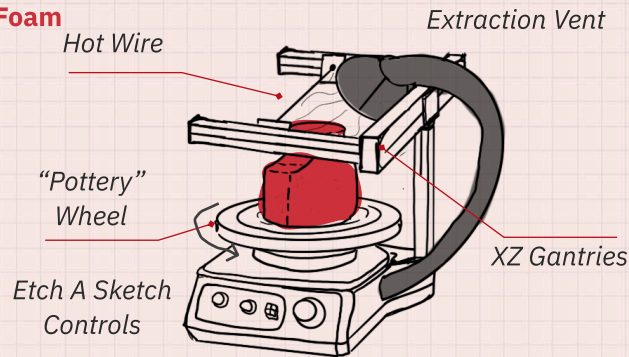
Ruaridh Murdoch

Creative Ideation from Insights

■ Cardboard ■ Foam

Vase Former

Foam

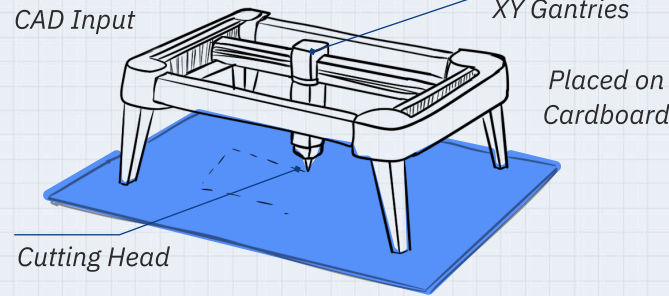


A foam shaping machine inspired by a pottery wheel.

The student mounts a foam block and adjusts the hot-wire cutter while the platform rotates. The platform can also rotate in steps to create geometric forms. An extraction vent removes fumes during cutting.

CAD Cutter

Cardboard

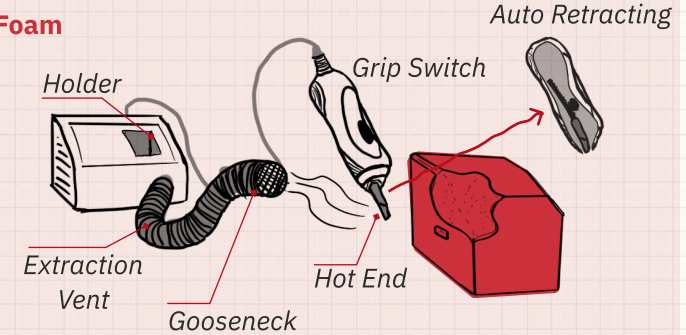


A lightweight CNC cutter designed for quick classroom use.

Students place the device on top of cardboard and send a simple design file to cut or score parts automatically. This allows accurate model components to be produced without manual knife work.

Safe Hot-End

Foam

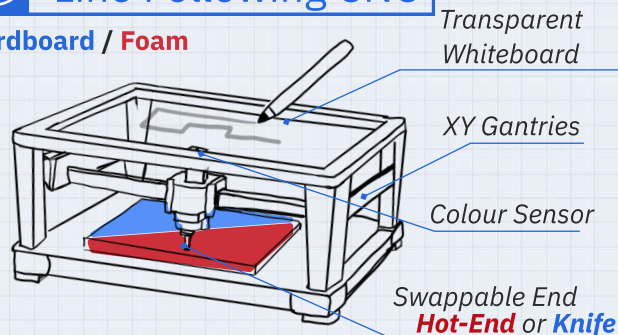


A handheld foam carving tool designed to make detailed shaping safer and easier.

The heated tip only extends while the grip is held, retracting when released. An extraction unit removes fumes and provides a safe place to store the tool when not in use.

Line Following CNC

Cardboard / Foam

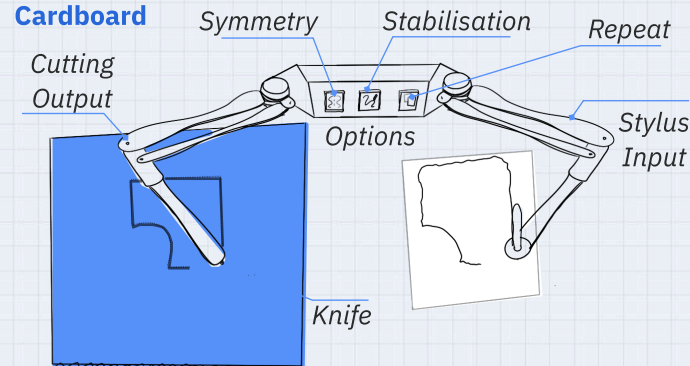


A drawing-guided cutting machine that follows lines sketched by the student.

Students draw their path on the whiteboard. A colour sensor follows the line and guides a motorised cutter to reproduce the shape in foam or cardboard below. No CAD or precise knife work required.

Perfect Pantograph

Cardboard

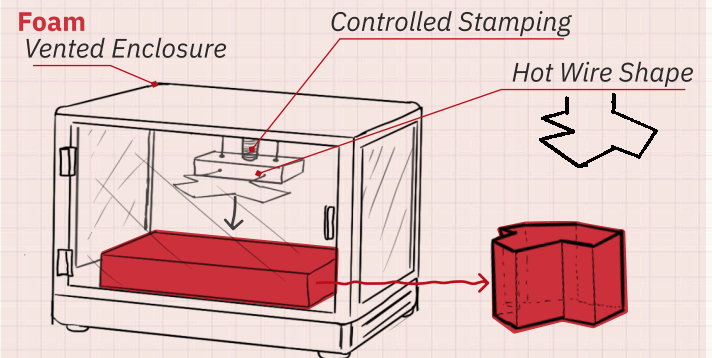


Portable system that turns drawings into precise cuts.

The student moves a stylus to sketch the shape they want, while a mirrored arm performs the cut in cardboard. Stabilisation, symmetry and repeat options help produce cleaner and more consistent parts.

Cookie Cutter

Foam



Cutting press that uses custom bent heated wire.

Students create their own wire outline and insert it into the machine before pressing it through the foam. This produces clean, repeatable shapes while containing fumes inside the enclosure.

Concept Selection

Feedback & Selection

Scoring Matrix

Analysed all six concepts using four criteria derived from the research and project constraints. Each concept was scored from 1-5 for each criteria, scores were then weighted and summed to produce a final ranking.

Criteria		Weighting
Impact	How much the concept improves students' ability to create accurate prototypes.	35%
Feasibility	How realistic the concept is to engineer and prototype.	30%
Safety	How safe the concept is for use in a DT workshop.	20%
Control	Whether the student still controls the design outcome.	15%

Concept	Impact	Feasibility	Safety	Control	Score
Vase Former	4	3	3	4	3.45
CAD Cutter	3	4	4	2	3.35
Safe Hot End	4	5	4	4	4.35
Line CNC	3	3	4	3	3.20
P. Pantograph	5	4	4	5	4.50
Cookie Cutter	3	3	3	2	2.84

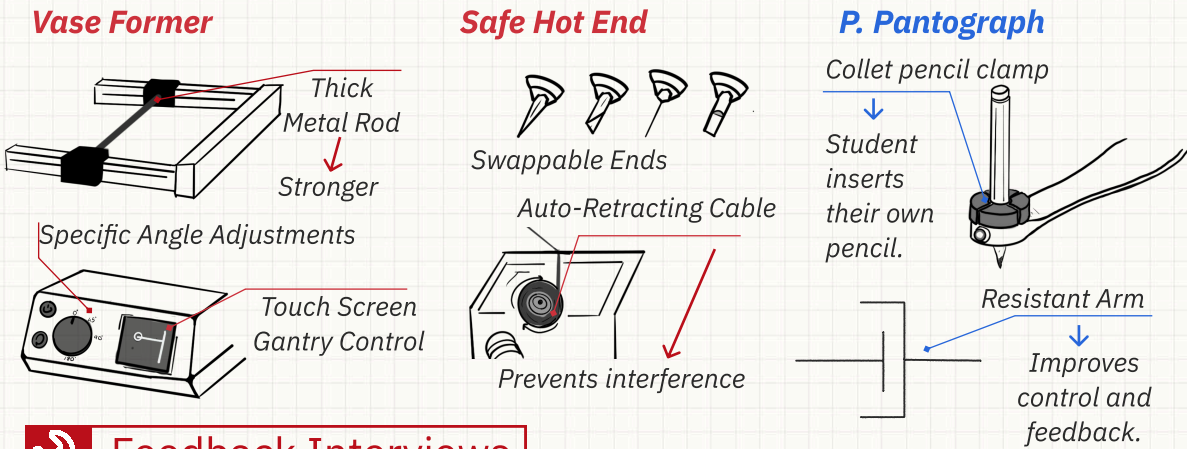
The three best-scoring concepts were selected for further development:

Vase Former **Safe Hot End** **P. Pantograph**

These showed the strongest overall balance and were taken forward for further development.

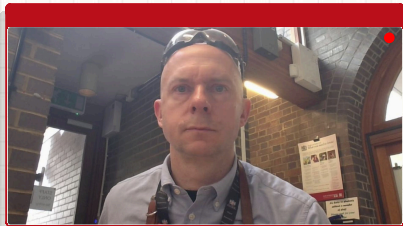
Ruaridh Murdoch

Further Development



Feedback Interviews

Short interviews were conducted with **Luke Barker** (Head of DT, City of London School for Girls) and two DT students from different schools to gather reactions to the three concepts.



Vase Former

Seen as useful but less exciting. Angle guidance (45° / 90°) could help students cut more accurately.

Safe Hot End

Students liked the idea of carving foam freely. Safety features were seen as helpful for preventing burns.

Concerns about fumes from melting foam. Foam modelling is also less common in schools due to cost, with cardboard used more often.

Perfect Pantograph

Most positive response. Drawing interaction felt intuitive and similar to tools students already use. Helpful for students who can sketch but struggle to cut accurately.

Based on the feedback and scoring matrix, the Perfect Pantograph was selected as the concept to take forward for further development.

Concept Development

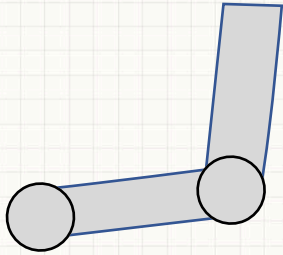
Ruaridh Murdoch

Perfect Pantograph Exploration

Arm

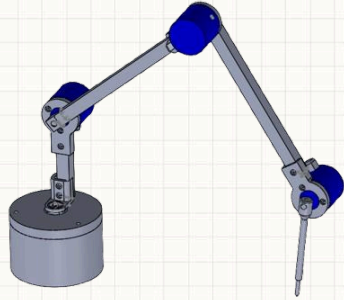
The input arm needs to track 2D position. Three approaches considered:

Planar Arm



Rotary potentiometers at joints. Simple and efficient.

CMM Digitiser Arm



Allows pen angle but adds joints, cost, and fragility

Camera Tracking



No arm needed, but lighting could make it unreliable

Planar arm with rotary potentiometers is preferred as it is sufficient accuracy for cardboard. The cutting arm mirrors the processed input in near real-time, both servos sat at the shoulder to reduce inertia and student tampering, moving the arm with linkages.



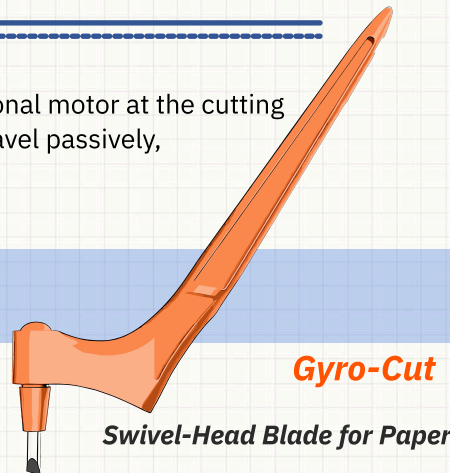
For blade lift, a third servo is needed for pressure control, allowing the device to both score and cut.

For curves, a fixed razor blade requires an additional motor at the cutting head. A swivel blade self-aligns to direction of travel passively, significantly simplifying the head.

Cutting footprint should cover:

A3
AREA

for versatile modelling

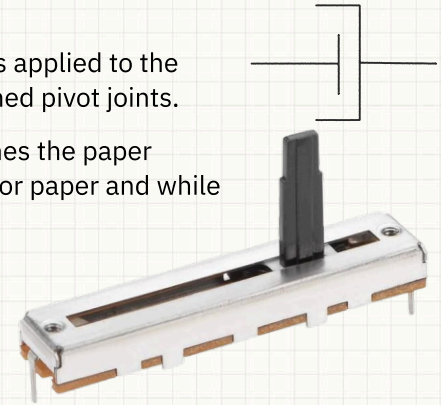


Inputs

To make movement feel deliberate, resistance is applied to the stylus arm via controlled friction through tightened pivot joints.

The system needs to detect when the pen touches the paper without adding weight or electronics to the pen or paper and while allowing different pen heights.

This could be done by measuring vertical movement using a linear potentiometer near the shoulder.



Functional modes, inspired by **Procreate's** input processing:

- **Stabilisation** — weighted moving average smooths input jitter before sending to cutting arm
- **Symmetry** — mirrors path across a defined axis
- **Axis-only** — constrains output to horizontal/vertical for straight cuts
- **Repeat** — stores and replays a path with defined spacing for batch cutting
- **Perfect Circle** — fits a circle to the input gesture

Materials & Manufacture

The housing should be **ABS**. It is impact resistant and easy to mould. The arms would need to be stiffer, flex could introduce path error. Thick aluminium sheet resists this while staying lightweight.

Design for Manufacture:

One fastener size throughout. Consistent wall thickness to avoid moulding problems and snap fit internals to reduce assembly (seen in the teardown).

Design for Assembly:

The interviews showed that students could not confidently change blades on machines like scroll saws, this created an urgent need for a teacher, slowing down workflow.

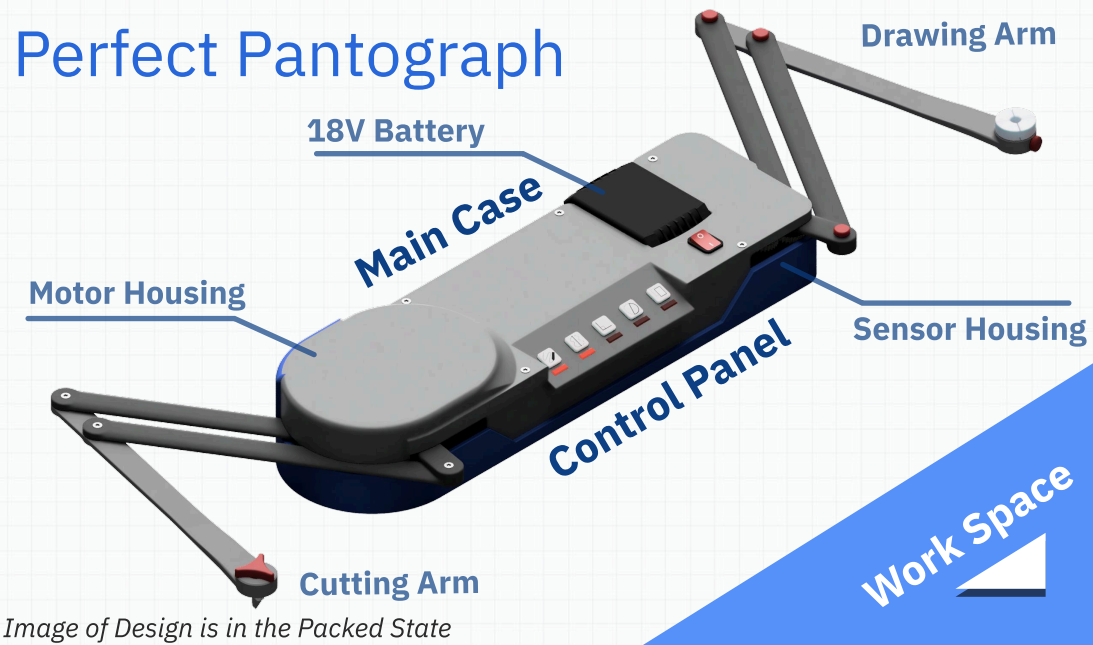
- *The blade should be easy to swap when it gets dull, having a locking mechanism that students understand.*
- *Electronics access should require a different tool so teachers can get in but students cannot.*

Technical Design

Ruaridh Murdoch

3D Model, Features and Rationale

Perfect Pantograph

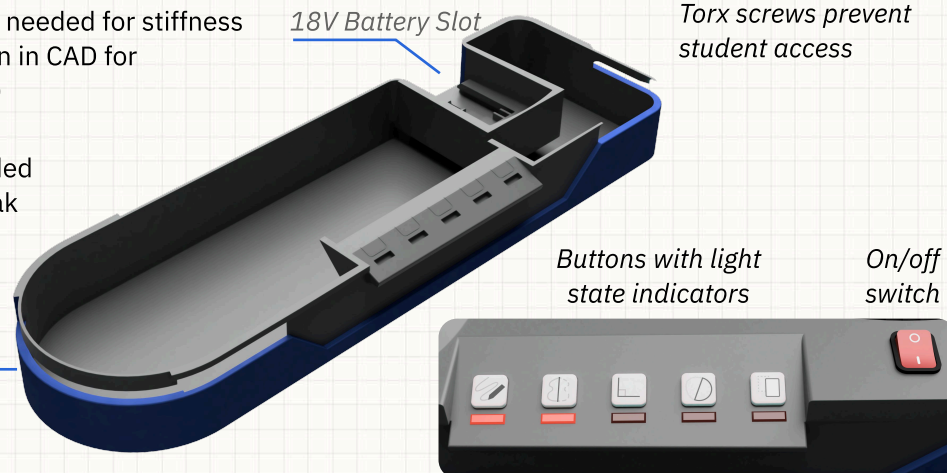


Case Design

ABS, 3 mm wall thickness for injection moulding

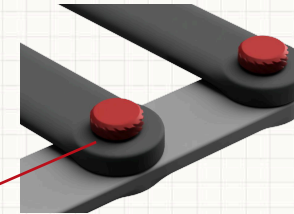
Ribs likely needed for stiffness (not shown in CAD for simplicity)

Overmoulded ABS in weak areas for strength + colour separation

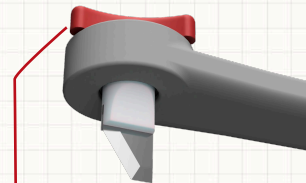
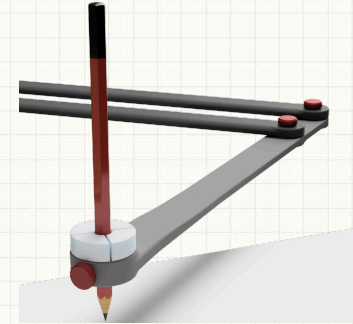


Arm Design

Knurled tighteners for adjustable resistance for control



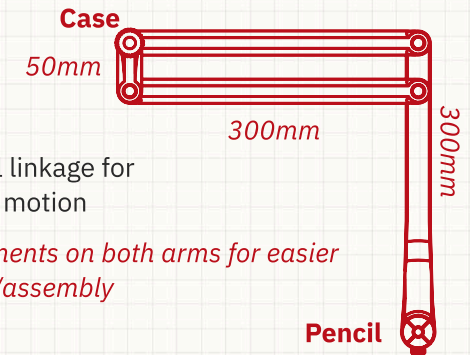
Collet clamp, student inserts own pencil



Swivel blade on bearing, screw top to remove blade system.

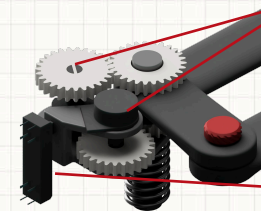
4-bar parallel linkage for stable planar motion

Same components on both arms for easier manufacture/assembly



Mechanisms

Drawing Arm Sensors



10kΩ Rotary Potentiometers

Offset pots using 1:1 gear linkage

Spring so natural arm position is up

250k Linear Slider Potentiometer

Cased together so it can be moved vertically

20mm vertical travel

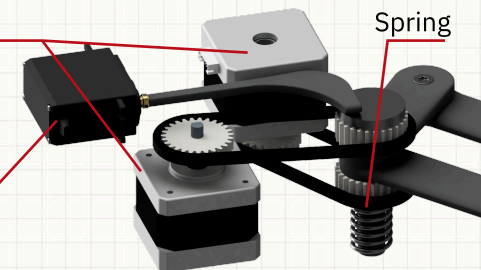
Cutting Arm Actuators

NEMA 17 Stepper Motors

For accurate and strong measurement
Linked using belts to allow vertical movement

MG996R Servo Motor

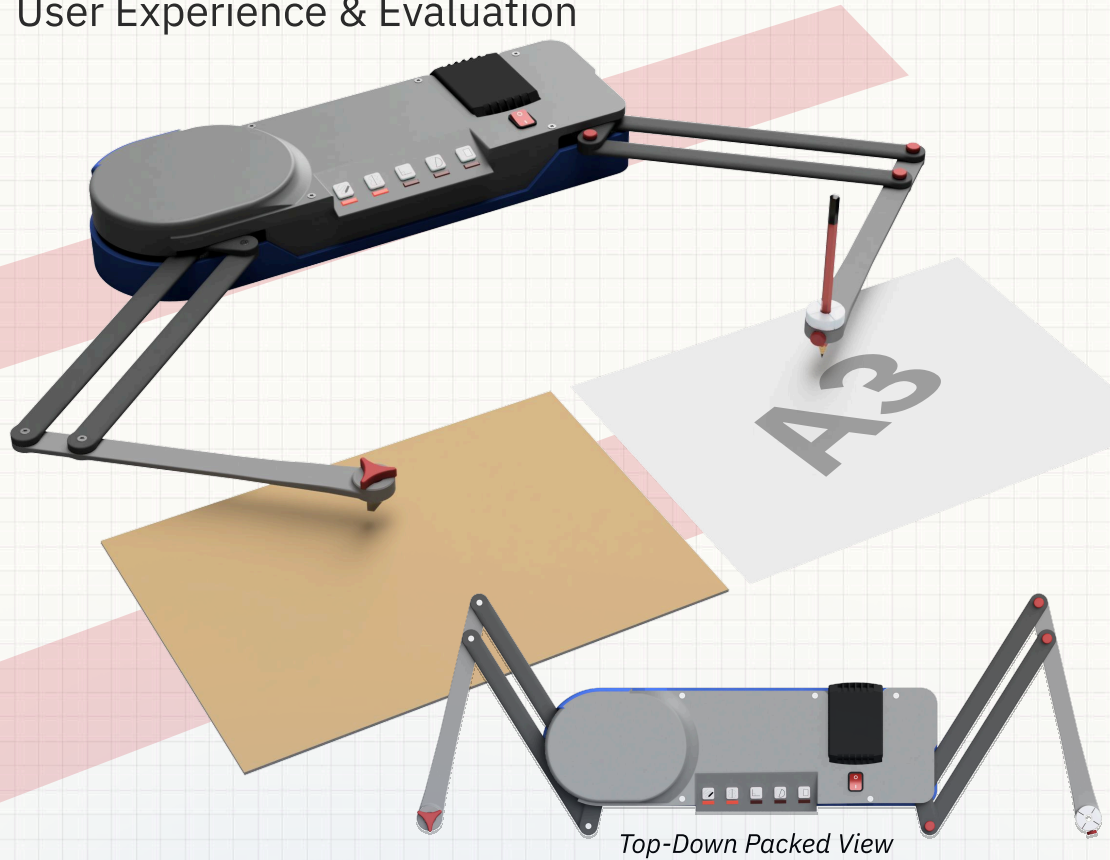
High torque metal geared servo for accurate blade pressure



Hard stops to prevent arm collision

Final Design

User Experience & Evaluation



Top-Down Packed View

Instructions for Use

1. Retrieve the tool from storage in its packed state.
2. Place it on the desk and insert the battery.
3. Insert a pencil, tighten the collet, and press it to the paper while turning the tool on (calibrates height).
4. The cutting arm now mirrors your actions. Pressing the pencil down engages the cut.
5. Trace an existing design or draw a new one.
6. Press any of the option buttons to perfect and refine. Press stabilise + repeat to score for folds.
7. When finished, remove the battery and pencil, then return the tool to storage.

You now have clean, accurate cardboard parts ready for modelling.

Ruaridh Murdoch



Product Description

The **Perfect Pantograph** is a desktop cutting tool that lets **DT students translate hand-drawn shapes into precise cardboard cuts**. The student sketches a path with the drawing arm while the cutting arm mirrors the movement in real time, moving a blade through cardboard with **consistent pressure and accuracy**. On-board processing **perfects the input before cutting**, so the output is always cleaner than the hand that drew it. Designed for shared school workshops, it requires **no CAD knowledge, no teacher setup or supervision, and no manual knife control**.

Stabilisation



Smooths shaky input so lines cut cleanly.

Symmetry



Mirrors your path across a central axis as you draw.

Axis Only



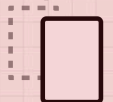
Locks movement to axis for straight cuts.

Perfect Circle



Fits your drawing to a clean circle automatically.

Repeat



Stores and replays the path again for batch cuts.



Evaluation

Students avoid modelling because cuts are inaccurate and take too long. When students trust that a cut will come out accurately, modelling becomes a better part of the design process rather than an obstacle to it.

Safe

No handheld blade reduces the risk of cuts and slips. The blade remains exposed during cutting however, so a blade guard is identified as a development priority.

Indepandant

No teacher setup or supervision required. Students can work continuously without waiting.

Quick and Simple

Fast setup, no CAD needed. Maximises making time in short lessons.

Reduce Manual Skill

Accuracy comes from the mechanism, not the user's hand.

Clean

A sharp blade cutting cardboard produces minimal debris, making workspace tidy.